# Aero Pass

**Designed and Developed by**


| | |
|---|---|
| T. ARUN KUMAR | 2311CS010645 |
| Y. SANHITH REDDY | 2311CS010691 |
| Y. ABHISHEK | 2311CS010693 |
| V. SIDDARTHA | 2311CS010688 |

**Guided by**

**Dr. B. Jogeswara Rao**

**Associate Professor**

## Department of Computer Science & Engineering

**MALLA REDDY UNIVERSITY, HYDERABAD**


**2024-2025**

MALLA REDDY UNIVERSITY

(Telangana State Private Universities Act No.13 of 2020 and G.O.Ms.No.14, Higher Education (UE) Department)

# CERTIFICATE

This is to certify that the Application development lab record entitled development lab record entitled **"Aero Pass"** submitted by **T.ARUN KUMAR (2311CS010645)**, **Y.SANHITH REDDY (2311CS010691), Y.ABHISHEK (2311CS010693), V.SIDDARTHA (2311CS010688)** B Tech II year I semester, Department of CSE during the year 2024-25. The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

**Internal Guide**

**Dr. B. Jogeswara Rao**

**(Associate Professor)**

**HOD-CSE**

**Dr.Meeravali Shaik**

**External Examiner**

MALLA REDDY UNIVERSITY
(Telangana State Private Universities Act No.13 of 2020 and G.O.Ms.No.14, Higher Education (UE) Department)

# DECLARATION

I declare that this project report titled **AERO PASS** submitted in partial fulfillment of the degree of B. Tech in CSE is a record of original work carried out by me under the supervision of **Dr. B. JOGESWARA RAO**, and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

| | |
|---|---|
| T. ARUN KUMAR | 2311CS010645 |
| Y. SANHITH REDDY | 2311CS010691 |
| Y. ABHISHEK | 2311CS010693 |
| V. SIDDARTHA | 2311CS010688 |

# ACKNOWLEDGEMENT

With great pleasure we want to take this opportunity to express our heartfelt gratitude to all the people who helped in making this mini project work a grand success.

We express our deep sense of gratitude to **Dr.B.JOGESWARA RAO** for his constant guidance throughout our mini project work.

We are grateful to **Mr.M.Rakesh** for his/her valuable suggestions and guidance given by him during the execution of this mini project work.

We would like to thank **Dr. Meeravali Shaik,** Head of the Department of Computer Science and Engineering, for providing seamless support and right Suggestions in the development of App.

We are highly indebted to **Vice Chancellor**, for giving us the permission to carry out this mini project.

Last but not the least we would like to thank the **Teaching & NonTeaching** staff of CSE Department for sharing their knowledge with us.

| | |
|---|---|
| T. ARUN KUMAR | 2311CS010645 |
| Y. SANHITH REDDY | 2311CS010691 |
| Y. ABHISHEK | 2311CS010693 |
| V. SIDDARTHA | 2311CS010688 |

# TABLE OF CONTENTS

# LIST OF FIGURES

# ABSTRACT

Aeropass is a cutting-edge airline reservation system designed to transform the way passengers book flights and manage their travel plans. This innovative platform provides an intuitive interface for users to search for flights, compare prices, and complete bookings securely. By streamlining the entire reservation process, Aeropass enhances user convenience and delivers a seamless experience, ensuring that travelers can plan their journeys with ease and efficiency.

A unique feature of Aeropass is its integrated parking reservation system, which allows passengers to book parking spaces for two-wheelers or four-wheelers alongside their flight tickets. This eliminates the hassle of finding parking at the airport, ensuring a stress-free start to their journey. By combining flight and parking reservations in one platform, Aeropass caters to the comprehensive needs of modern travelers.

To further enhance user satisfaction, Aeropass introduces Aero Assistance, an intelligent virtual assistant designed to address all user queries about the system. Whether customers need help understanding booking features, flight policies, or parking options, Aero Assistance provides quick and accurate responses. This feature ensures users can navigate the platform confidently and access all its functionalities without confusion.

Additionally, Aeropass offers an exclusive benefit for parking slot bookings: free luggage transportation from the parking area to the flight. This thoughtful feature ensures a hassle-free experience for travelers, allowing them to move comfortably through the airport without worrying about carrying heavy luggage. It reflects Aeropass commitment to delivering convenience at every step of the journey.

Aeropass stands out by addressing both passenger and airline needs through its innovative features, seamless integrations, and user-centric design. By combining flight booking, parking reservations, Aero Assistance, and complimentary luggage services, Aeropass redefines convenience and sets a new benchmark in the travel industry

# CHAPTER – 1

# INTRODUCTION

## 1.1 Introduction

The airline industry has seen significant growth over the past few decades, and as the demand for air travel continues to rise, so does the need for efficient and user-friendly reservation systems. Traditional booking methods, whether through travel agents or phone-based systems, often involve cumbersome processes and delays. Aeropass aims to address these challenges by offering a modern, fully integrated airline reservation platform that simplifies the flight booking process for passengers and streamlines the operational management for airlines. By harnessing the power of technology, Aeropass offers a seamless, secure, and intuitive experience for users.

At its core, Aeropass is designed to enhance the user experience by providing a user-friendly interface that allows passengers to search for flights based on various criteria such as destination, travel dates, and class. The system offers real-time availability and dynamic pricing, which ensures that passengers have access to up-to-date flight information and can make informed decisions when booking their travel. Additionally, Aeropass allows users to customize their bookings by selecting preferred seats, managing payment options, and even making modifications to their travel plans with ease. With these capabilities, Aeropass is not only a tool for booking flights.

For airlines, Aeropass provides a robust backend system that allows operators to efficiently manage bookings, monitor flight occupancy, and access detailed reports on booking trends and revenue. The platform is designed to integrate seamlessly with existing airline databases, offering features such as automated seat assignments, real-time flight updates, and customer management tools. With Aeropass, airlines can enhance operational efficiency, reduce manual work, and provide better service to their customers. The system’s flexibility and scalability make it suitable for a wide range of airline sizes, from smaller regional carriers to large international airlines, ultimately creating a mutually beneficial environment for both the passengers and the airlines.

## 1.2 Problem Statement

The airline reservation process, both for passengers and airline operators, has traditionally been plagued by inefficiencies, delays, and complexities. Passengers often face challenges when searching for available flights, comparing prices, or navigating through multiple platforms to complete their bookings. The lack of a streamlined, user-friendly interface in many legacy systems makes it time-consuming and frustrating for customers to make informed travel decisions. Moreover, passengers may encounter difficulties with seat selection, cancellations, and modifications to their reservations, which adds to their overall travel frustration.

Aeropass addresses these problems by offering a modern, integrated solution that simplifies the flight booking experience for passengers and streamlines reservation management for airlines. By providing real-time flight data, seamless booking processes, and comprehensive backend tools for operators, Aeropass aims to reduce booking errors, improve operational efficiency, and enhance customer satisfaction. This system ensures that both passengers and airline staff can manage reservations effortlessly, providing a smooth and efficient experience for all parties involved.

For airlines, managing reservations manually or through outdated systems can lead to errors, inefficiencies, and resource mismanagement. Many airlines still rely on cumbersome legacy systems or disconnected software that do not provide real-time updates on flight availability, pricing, or booking statuses. This not only increases the administrative burden but also leads to higher operational costs and a reduced ability to respond quickly to market demands. Additionally, airlines often face difficulties in tracking customer preferences, managing cancellations, and handling last-minute bookings, all of which can negatively impact customer satisfaction and the bottom line.

## 1.3 Objective

The primary objective of the Aeropass airline reservation system is to provide an intuitive, efficient, and reliable platform that simplifies the process of booking flights for passengers while optimizing flight management and reservation tracking for airlines. The system aims to address key challenges within the airline reservation process and enhance the overall user experience. The specific objectives of the Aeropass system include:

1. Simplified Flight Booking: Provide an easy-to-use interface for passengers to search flights, compare prices, and book tickets quickly.

2. Secure Payment System: Enable safe and seamless online payments with multiple payment options and instant booking confirmation.

3. Airline Management Tools: Offer airlines a backend system to efficiently manage flight schedules, seat occupancy, and reservations.

4. Customer-Focused Features: Include seat selection, booking modifications, and personalized recommendations for an enhanced user experience.

5. Real-Time Data Integration: Ensure up-to-date flight information, pricing, and seat availability to reduce errors and improve decision-making.

6. Scalability and Security: Build a scalable system with high performance and robust security for user data and transactions.

## 1.4 Goal of project

The primary goal of the Aeropass project is to develop a robust and user-friendly airline reservation system that simplifies the process of flight booking for passengers and enhances the operational efficiency of airlines. By providing a seamless, secure, and intuitive platform, Aeropass aims to offer a solution that meets the growing demands of the airline industry and improves the overall travel experience for users.

For passengers, the goal is to create an easily navigable system that allows for quick flight searches, secure payments, and straightforward booking management, while offering additional features like seat selection and real-time flight updates. This aims to reduce booking time, minimize user frustration, and provide a personalized, efficient service.

For airlines, the goal is to provide an integrated backend system that streamlines reservation tracking, flight management, and data reporting, ultimately improving operational workflows and customer service. The system will empower airlines to handle bookings more effectively, respond to demand shifts in real-time, and manage flight information and seat availability with greater ease and accuracy.

Furthermore, Aeropass aims to improve security and data privacy by incorporating encryption and compliance with global standards for financial transactions and user data protection. The system will also be designed to scale easily to handle large volumes of traffic, ensuring consistent performance during peak travel periods.

Another key goal is to support the airline industry's digital transformation by offering a fully integrated system that can be easily adapted to work with existing infrastructure. Aeropass seeks to minimize system downtime and enhance customer service by offering real-time flight updates, automated booking confirmations, and rapid responses to booking inquiries.

# CHAPTER – 2

# Problem Identification

## 2.1 Existing System

The current airline reservation systems in use by most airlines are often outdated, disconnected, or overly complex, leading to inefficiencies both for passengers and airline operators.  These legacy systems typically rely on a combination of manual processes, fragmented software tools, and limited automation, resulting in several challenges.

For passengers, existing systems can be cumbersome and difficult to navigate. Many airline websites and booking platforms feature complex interfaces, requiring users to input multiple details manually, which can be time-consuming and prone to error. Moreover, most systems do not provide real-time updates on flight availability or pricing, which can lead to confusion and frustration. Passengers may also face challenges when modifying or canceling their bookings, as these processes often require additional customer support intervention, resulting in delays and dissatisfaction.

For airline operators, the existing reservation systems often lack integration with other crucial systems, such as flight scheduling, customer management, and payment processing. This disjointed approach means that airlines may struggle with managing large volumes of bookings, tracking seat occupancy, and responding quickly to market demand. Furthermore, many airlines still rely on manual intervention to handle reservation changes, cancellations, and overbookings, which increases the risk of human error and operational inefficiencies. As a result, these systems may lead to lower customer satisfaction, reduced operational efficiency, and higher costs.

In summary, the existing airline reservation systems are often inadequate for the needs of modern airlines and passengers. They fail to provide seamless, real-time interactions and require significant manual effort from both users and airline staff. As a result, there is a clear need for a more efficient, user-friendly, and automated solution to address these challenges.

## 2.1 Proposed System

The Aeropass airline reservation system is designed to address the limitations of traditional reservation platforms by offering a modern, integrated, and user-friendly solution for both passengers and airline operators. The proposed system leverages advanced technology to automate key aspects of the booking process, enhance user experience, and streamline airline operations, ensuring efficiency and convenience for both travelers and airlines.

For passengers, Aeropass offers a seamless interface that allows users to search for flights, compare prices, and complete bookings with ease. The system supports various filters, such as destination, travel dates, flight class, and seat preferences, enabling users to quickly find options that suit their needs. Beyond booking flights, Aeropass introduces a parking slot reservation feature, allowing passengers to reserve parking spaces for two-wheelers and four-wheelers at the airport. This additional functionality ensures a hassle-free travel experience by eliminating the stress of last-minute parking arrangements.

Aeropass also includes Aero Chat Assistance, a smart virtual assistant that helps users navigate the platform and understand its features. Whether travelers have questions about flight booking, parking reservations, or general usage, Aero Chat Assistance provides instant and accurate responses. This user-friendly feature enhances accessibility and ensures that even first-time users can interact with the system confidently.

An exclusive benefit offered by Aeropass is free luggage transportation support for passengers who book parking slots through the platform. This thoughtful feature ensures that travelers can move effortlessly from their parked vehicle to their flight, with their luggage handled professionally. By integrating parking and luggage assistance, Aeropass delivers a comprehensive service that prioritizes passenger convenience.

For airlines, the system provides a robust backend for managing flight schedules, seat availability, and booking data in real time. Aeropass integrates seamlessly with other airline systems, such as payment gateways and customer management platforms, creating a unified operational framework. With advanced reporting tools and real-time analytics, airlines can optimize their pricing strategies, identify trends, and improve decision-making, ensuring better alignment with market demands.

# CHAPTER – 3
# Requirements

## 3.1 Software Requirements

FRONT END : Java (Swing Components)

BACK END : Java

OPEERATING SYSTEM: Windows 11

IDE : Netbeans

Data base : mysql

## 3.2 Hardware Requirements Database Server:

i. Processor: Similar to the application server, a multi-core processor for handling database queries efficiently.

ii. Storage: SSD storage is preferable for faster data retrieval.

iii. RAM: Sufficient RAM for caching and handling database operations. Again, the amount depends on the size of the database.

# CHAPTER – 4

# Design and Implementation

## 1.1Design

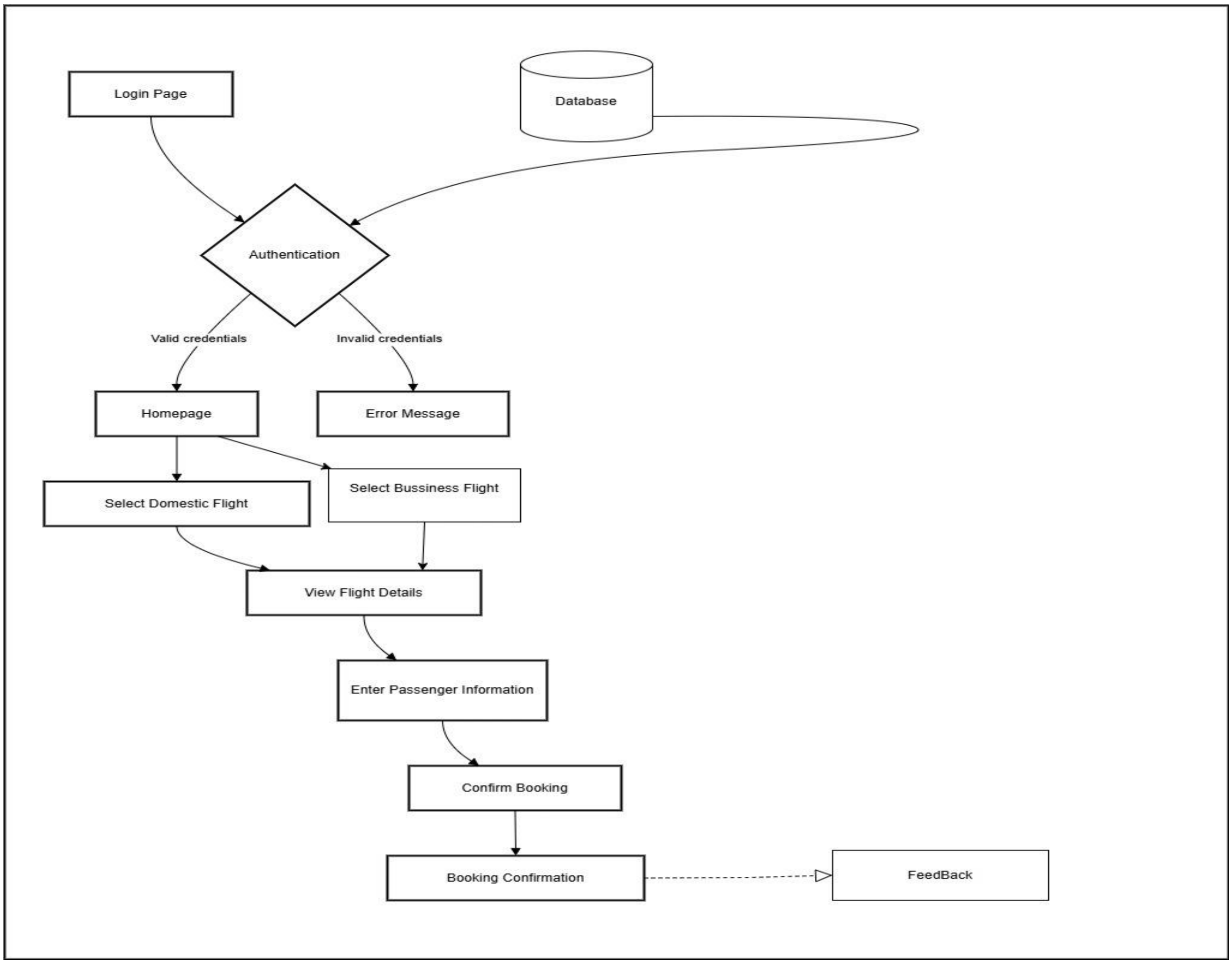
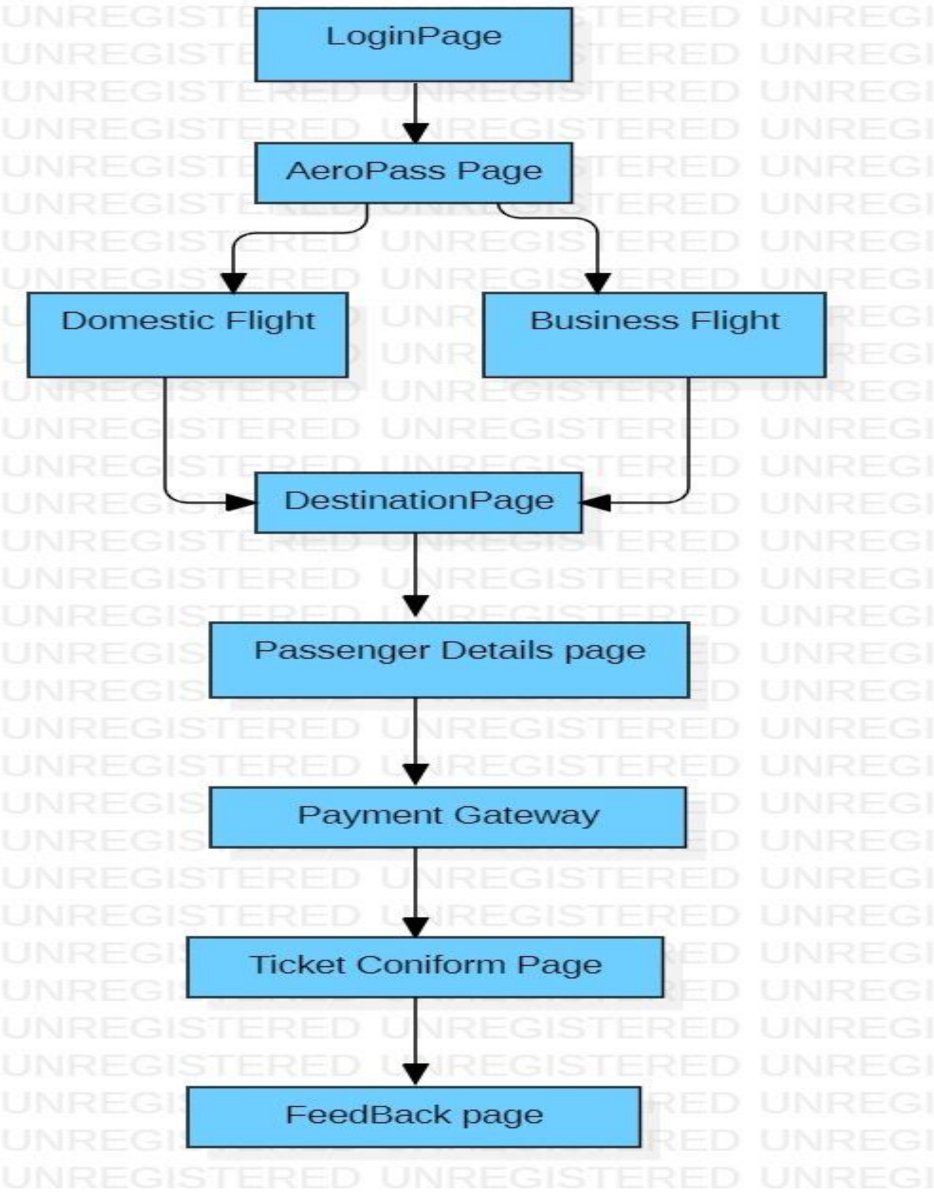


**Fig 1.2 : Data Flow level Diagram**



## 4.1Implementation.

The Aeropass airline reservation system will be implemented using Java for both the back-end and integration with front-end technologies, databases, and payment systems, ensuring scalability, security, and optimal performance.

**Swing GUI Implementation**

The user interface for the **Aeropass Airline Reservation System** is built using **Java Swing** components. The system includes various windows and forms for user interaction, such as:

**Login Window**: Users enter their credentials to log in.

**Flight Search Window**: Users can search for flights based on source, destination, and date.

**Booking Window**: After selecting a flight, users input passenger details and confirm the booking.

**Admin Panel**: Admins manage flights, view reservations, and generate reports.

**Database Connectivity (JDBC)**

The system connects to a **MySQL** database using **JDBC** for all data operations. This includes fetching available flights, making reservations, and updating flight availability. SQL queries are executed using **PreparedStatement** to ensure secure and efficient data interaction.

**User and Admin Roles**

The system supports **role-based access control** where customers can book flights, and admins can manage flights and view system reports. This is controlled based on the **Users** table, which stores user roles.

**Database Structure**

The **Aeropass** system uses multiple tables in the MySQL database:

**Users Table**: Stores user credentials and roles.

**Flights Table**: Stores flight details such as flight number, source, and available seats.

**Reservations Table**: Tracks reservations, including user ID, flight ID, and booking status.

**Payments Table**: Records payment transactions associated with reservations.

# CHAPTER – 5
# CODE

## 5.1 Source Code

**LoginPage.java**

```
import DatabaseConnection.DatabaseConnection;

import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;

public class LoginPage extends JFrame {

public LoginPage() {
setTitle("Aeropass Login");
setSize(400, 500);  // Increased size to accommodate the logo
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLocationRelativeTo(null);

JPanel panel = new JPanel(new GridBagLayout());
panel.setBackground(Color.WHITE);
GridBagConstraints gbc = new GridBagConstraints();
gbc.insets = new Insets(5, 5, 5, 5);

// Load and resize Logo Image
JLabel logoLabel = new JLabel();
try {
ImageIcon logoIcon = new ImageIcon(getClass().getResource("/images/flight.jpeg"));
Image logoImage = logoIcon.getImage().getScaledInstance(350, 250, Image.SCALE_SMOOTH);
logoLabel.setIcon(new ImageIcon(logoImage));
} catch (Exception e) {
System.out.println("Logo image not found.");
}

gbc.gridx = 0;
gbc.gridy = 0;
gbc.gridwidth = 2;
panel.add(logoLabel, gbc);

// Login Components
JLabel titleLabel = new JLabel("Login");
titleLabel.setFont(new Font("Arial", Font.BOLD, 18));
titleLabel.setForeground(Color.BLUE);
```

```
gbc.gridy = 1;
panel.add(titleLabel, gbc);

JLabel emailLabel = new JLabel("Email:");
JTextField emailField = new JTextField(15);
JLabel passwordLabel = new JLabel("Password:");
JPasswordField passwordField = new JPasswordField(15);

JButton loginButton = new JButton("Login");
loginButton.setBackground(Color.BLUE);
loginButton.setForeground(Color.WHITE);

JLabel registerLabel = new JLabel("<HTML><U>Don't have an account? Register here</U></HTML>");
registerLabel.setForeground(Color.BLUE);
registerLabel.setCursor(Cursor.getPredefinedCursor(Cursor.HAND_CURSOR));

gbc.gridwidth = 1;
gbc.gridx = 0;
gbc.gridy = 2;
panel.add(emailLabel, gbc);
gbc.gridx = 1;
panel.add(emailField, gbc);

gbc.gridy = 3;
gbc.gridx = 0;
panel.add(passwordLabel, gbc);
gbc.gridx = 1;
panel.add(passwordField, gbc);

gbc.gridy = 4;
gbc.gridwidth = 2;
panel.add(loginButton, gbc);

gbc.gridy = 5;
panel.add(registerLabel, gbc);

add(panel);

// Action for login button
loginButton.addActionListener(e -> {
String email = emailField.getText();
String password = new String(passwordField.getPassword());

if (!email.isEmpty() && !password.isEmpty()) {
if (validateLogin(email, password)) {
JOptionPane.showMessageDialog(this, "Login Successful!", "Success", JOptionPane.INFORMATION_MESSAGE);
```

```
dispose();
new MainMenu().setVisible(true);  // Open the main menu
} else {
JOptionPane.showMessageDialog(this, "Invalid email or password", "Error", JOptionPane.ERROR_MESSAGE);
}
} else {
JOptionPane.showMessageDialog(this, "Please enter both email and password", "Error", JOptionPane.ERROR_MESSAGE);
}
});

// Action for register label (redirect to registration page)
registerLabel.addMouseListener(new MouseAdapter() {
@Override
public void mouseClicked(MouseEvent e) {
dispose();
new RegistrationPage().setVisible(true);  // Open the registration page
}
});

setVisible(true);
}

// Validate login using database credentials
private boolean validateLogin(String email, String password) {
String sql = "SELECT * FROM users WHERE email = ? AND password = ?";
try (Connection conn = DatabaseConnection.getConnection();
PreparedStatement pstmt = conn.prepareStatement(sql)) {

pstmt.setString(1, email);
pstmt.setString(2, password);
ResultSet rs = pstmt.executeQuery();

return rs.next();  // Return true if login successful

} catch (SQLException e) {
System.out.println("Database error: " + e.getMessage());
JOptionPane.showMessageDialog(this, "Database error. Please try again later.", "Error", JOptionPane.ERROR_MESSAGE);
return false;
}
}

public static void main(String[] args) {
SwingUtilities.invokeLater(LoginPage::new);
}
}
```

**Regestration.java**

```
import DatabaseConnection.DatabaseConnection;
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.net.URL;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.SQLException;

public class RegistrationPage extends JFrame {
public RegistrationPage() {
setTitle("AeroPass Registration");
setSize(700, 400);  // Adjusted the size for better layout
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLocationRelativeTo(null);

// Set light blue background color for the frame
getContentPane().setBackground(new Color(173, 216, 230)); // Light blue color

// Panel for registration components
JPanel registrationPanel = new JPanel(new GridBagLayout());
registrationPanel.setOpaque(false); // Transparent to show frame background
GridBagConstraints gbc = new GridBagConstraints();
gbc.insets = new Insets(5, 5, 5, 5);
gbc.fill = GridBagConstraints.HORIZONTAL;

JLabel titleLabel = new JLabel("Aeropass Registration");
titleLabel.setFont(new Font("Arial", Font.BOLD, 18));
titleLabel.setForeground(Color.BLUE);
gbc.gridx = 0;
gbc.gridy = 0;
gbc.gridwidth = 2;
registrationPanel.add(titleLabel, gbc);

JLabel usernameLabel = new JLabel("Username:");
JTextField usernameField = new JTextField(15);
JLabel nameLabel = new JLabel("Name:");
JTextField nameField = new JTextField(15);
JLabel emailLabel = new JLabel("Email:");
JTextField emailField = new JTextField(15);
JLabel passwordLabel = new JLabel("Password:");
JPasswordField passwordField = new JPasswordField(15);
JLabel confirmPasswordLabel = new JLabel("Confirm Password:");
JPasswordField confirmPasswordField = new JPasswordField(15);

JButton registerButton = new JButton("Register");
```

```
JButton backButton = new JButton("Back");  // Back button to return to LoginPage

gbc.gridwidth = 1;
gbc.gridx = 0;
gbc.gridy = 1;
registrationPanel.add(usernameLabel, gbc);
gbc.gridx = 1;
registrationPanel.add(usernameField, gbc);

gbc.gridy = 2;
gbc.gridx = 0;
registrationPanel.add(nameLabel, gbc);
gbc.gridx = 1;
registrationPanel.add(nameField, gbc);

gbc.gridy = 3;
gbc.gridx = 0;
registrationPanel.add(emailLabel, gbc);
gbc.gridx = 1;
registrationPanel.add(emailField, gbc);

gbc.gridy = 4;
gbc.gridx = 0;
registrationPanel.add(passwordLabel, gbc);
gbc.gridx = 1;
registrationPanel.add(passwordField, gbc);

gbc.gridy = 5;
gbc.gridx = 0;
registrationPanel.add(confirmPasswordLabel, gbc);
gbc.gridx = 1;
registrationPanel.add(confirmPasswordField, gbc);

// Panel to hold both "Back" and "Register" buttons side by side
JPanel buttonPanel = new JPanel(new FlowLayout(FlowLayout.CENTER, 10, 0));
buttonPanel.add(backButton);
buttonPanel.add(registerButton);

gbc.gridy = 6;
gbc.gridwidth = 2;
registrationPanel.add(buttonPanel, gbc);

// Logo Panel with Image from the Web
JPanel logoPanel = new JPanel(new BorderLayout());
logoPanel.setOpaque(false); // Transparent to show frame background
try {
URL logoURL = new
URL("https://tse3.mm.bing.net/th?id=OIP.ONMbEpR0fKNzjLgOXScpQAAAAA&pid=Api&P=0&h
```

```
=180"); // Replace with the URL of your logo
ImageIcon logoIcon = new ImageIcon(logoURL);
JLabel logoLabel = new JLabel(logoIcon);
logoLabel.setHorizontalAlignment(SwingConstants.CENTER);
logoPanel.add(logoLabel, BorderLayout.CENTER);
} catch (Exception e) {
e.printStackTrace();
}

// Main panel combining registration and logo panels with padding for better spacing
JPanel mainPanel = new JPanel(new GridBagLayout());
mainPanel.setOpaque(false); // Transparent to show frame background
GridBagConstraints mainGbc = new GridBagConstraints();
mainGbc.insets = new Insets(20, 20, 20, 20); // Adjust padding for spacing between components
mainGbc.gridx = 0;
mainGbc.gridy = 0;
mainPanel.add(registrationPanel, mainGbc);
mainGbc.gridx = 1;
mainPanel.add(logoPanel, mainGbc);

add(mainPanel);

registerButton.addActionListener(e -> {
String username = usernameField.getText();
String name = nameField.getText();
String email = emailField.getText();
String password = new String(passwordField.getPassword());
String confirmPassword = new String(confirmPasswordField.getPassword());

if (username.isEmpty() || name.isEmpty() || email.isEmpty() || password.isEmpty() ||
confirmPassword.isEmpty()) {
JOptionPane.showMessageDialog(this, "Please fill in all fields", "Error",
JOptionPane.ERROR_MESSAGE);
} else if (!password.equals(confirmPassword)) {
JOptionPane.showMessageDialog(this, "Passwords do not match", "Error",
JOptionPane.ERROR_MESSAGE);
} else {
if (registerUser(username, name, email, password)) {
JOptionPane.showMessageDialog(this, "Registration Successful!", "Success",
JOptionPane.INFORMATION_MESSAGE);
dispose();
new LoginPage().setVisible(true);
} else {
JOptionPane.showMessageDialog(this, "Registration Failed. Please try again.", "Error",
JOptionPane.ERROR_MESSAGE);
}
}
});
```

```
backButton.addActionListener(e -> {
dispose(); // Close the registration page
new LoginPage().setVisible(true); // Open the login page
});

setVisible(true);
}

private boolean registerUser(String username, String name, String email, String password) {
String sql = "INSERT INTO users (username, name, email, password) VALUES (?, ?, ?, ?)";

try (Connection conn = DatabaseConnection.getConnection();
PreparedStatement pstmt = conn.prepareStatement(sql)) {

pstmt.setString(1, username);
pstmt.setString(2, name);
pstmt.setString(3, email);
pstmt.setString(4, password);
pstmt.executeUpdate();
return true;
} catch (SQLException e) {
e.printStackTrace();
return false;
}
}

public static void main(String[] args) {
SwingUtilities.invokeLater(RegistrationPage::new);
}
}
```

**MainMenu.java**

```
import javax.swing.*;
import java.awt.*;

public class MainMenu extends JFrame {

public MainMenu() {
setTitle("Welcome to Aeropass");
setSize(400, 300); // Set size of the Main Menu window
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLocationRelativeTo(null); // Center the window on the screen

// Set light blue background color
JPanel panel = new JPanel();
panel.setBackground(new Color(173, 216, 230)); // Light blue color
```

```
setLocationRelativeTo(null); // Center the window on the screen

// Create the custom menu bar with back arrow icon
JMenuBar menuBar = new JMenuBar();
JMenu menu = new JMenu(); // Empty menu to hold the back arrow
menuBar.add(menu); // Add empty menu to the left side

// Load back arrow icon from URL
try {
URL arrowUrl = new
URL("https://upload.wikimedia.org/wikipedia/commons/thumb/e/e3/Back_Arrow.svg/32px-
Back_Arrow.svg.png");
ImageIcon backArrowIcon = new ImageIcon(new
ImageIcon(arrowUrl).getImage().getScaledInstance(16, 16, Image.SCALE_SMOOTH));
JLabel backArrowLabel = new JLabel(backArrowIcon);

backArrowLabel.setCursor(new Cursor(Cursor.HAND_CURSOR));
backArrowLabel.setToolTipText("Back to Main Menu");

// Add click event to the back arrow label
backArrowLabel.addMouseListener(new MouseAdapter() {
@Override
public void mouseClicked(MouseEvent e) {
dispose(); // Close this window
new MainMenu(); // Open MainMenu
}
});

menu.add(backArrowLabel); // Add back arrow to the menu
} catch (Exception e) {
e.printStackTrace();
}

setJMenuBar(menuBar); // Set custom menu bar as frame's menu bar

// Set up the main panel
JPanel panel = new JPanel(new BorderLayout());
panel.setBackground(Color.WHITE);

// Chat area to display conversation
chatArea = new JTextArea();
chatArea.setEditable(false);
chatArea.setLineWrap(true);
chatArea.setWrapStyleWord(true);
JScrollPane scrollPane = new JScrollPane(chatArea);

// Input field for user questions
inputField = new JTextField();
```

```
inputField.addActionListener(new SendMessageListener());

// Send button
JButton sendButton = new JButton("Send");
sendButton.addActionListener(new SendMessageListener());

// Back button
JButton backButton = new JButton("Back");
backButton.addActionListener(e -> {
new MainMenu();
dispose();
});

// Input panel with input field, back button and send button
JPanel inputPanel = new JPanel(new BorderLayout());
inputPanel.add(inputField, BorderLayout.CENTER);

JPanel buttonPanel = new JPanel(new FlowLayout(FlowLayout.RIGHT));
buttonPanel.add(backButton);
buttonPanel.add(sendButton);

inputPanel.add(buttonPanel, BorderLayout.EAST);

// Initialize conversation history manager
conversationHistory = new StringBuilder();

// Add components to the main panel
panel.add(scrollPane, BorderLayout.CENTER);
panel.add(inputPanel, BorderLayout.SOUTH);

add(panel);
setVisible(true);
}

// Listener class for handling user messages and bot responses
private class SendMessageListener implements ActionListener {
public void actionPerformed(ActionEvent e) {
String userMessage = inputField.getText().trim();
if (!userMessage.isEmpty()) {
addMessage("You: " + userMessage);
inputField.setText("");
String botResponse = getBotResponse(userMessage);
addMessage("Aero: " + botResponse);
}
}
}

// Method to add messages to the chat area and handle message count
```

```
public class AboutUsPage extends JFrame {

public AboutUsPage() {
setTitle("About Us");
setSize(400, 300); // Set size of the About Us window
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLocationRelativeTo(null); // Center the window on the screen

JPanel panel = new JPanel();
panel.setLayout(new BorderLayout());

// HTML description about Aeropass
JLabel aboutLabel = new JLabel("<html><center><h2>About Aeropass</h2>"
+ "<p>Aeropass is a premium loyalty program offering exclusive travel benefits.</p>"
+ "<p>Basically, Aeropass offers two flights: Domestic and Business flights, and it offers parking
booking as well.</p>"
+ "<p>Members enjoy priority boarding, discounted fares, and more.</p></center></html>");
aboutLabel.setHorizontalAlignment(SwingConstants.CENTER);
panel.add(aboutLabel, BorderLayout.NORTH);

// Social media panel
JPanel socialMediaPanel = new JPanel();

// Instagram icon and link
JLabel instagramLabel =
createSocialMediaLabel("https://upload.wikimedia.org/wikipedia/commons/thumb/9/95/Instagram_lo
go_2022.svg/1200px-Instagram_logo_2022.svg.png", "https://www.instagram.com/codinghub713/");
socialMediaPanel.add(instagramLabel);

// Twitter icon and link
JLabel twitterLabel = createSocialMediaLabel("https://cdn.punchng.com/wp-
content/uploads/2023/07/24084806/Twitter-new-logo.jpeg", "https://www.twitter.com");
socialMediaPanel.add(twitterLabel);

// Facebook icon and link
JLabel facebookLabel =
createSocialMediaLabel("https://upload.wikimedia.org/wikipedia/commons/thumb/f/ff/Facebook_log
o_36x36.svg/1200px-Facebook_logo_36x36.svg.png", "https://www.facebook.com");
socialMediaPanel.add(facebookLabel);

// Snapchat icon and link
JLabel snapchatLabel =
createSocialMediaLabel("https://th.bing.com/th/id/OIP.ZXD0bFXkQ0zescoxr-
Vj1AAAAA?rs=1&pid=ImgDetMain", "https://accounts.snapchat.com/v2/welcome");
socialMediaPanel.add(snapchatLabel);

panel.add(socialMediaPanel, BorderLayout.CENTER);
```

```
// Back button
JButton backButton = new JButton("Back");
backButton.addActionListener(e -> {
new MainMenu();
dispose();
});

JPanel buttonPanel = new JPanel(new FlowLayout());
buttonPanel.add(backButton);
panel.add(buttonPanel, BorderLayout.SOUTH);

add(panel);
setVisible(true);
}

private JLabel createSocialMediaLabel(String imageUrl, String link) {
// Create image icon from URL with larger size
JLabel label = new JLabel();
try {
URL url = new URL(imageUrl);
ImageIcon icon = new ImageIcon(new ImageIcon(url).getImage().getScaledInstance(50, 50,
Image.SCALE_SMOOTH)); // Increased size to 50x50
label.setIcon(icon);
} catch (Exception ex) {
ex.printStackTrace();
}

// Add click event to open link
label.setCursor(new Cursor(Cursor.HAND_CURSOR));
label.addMouseListener(new MouseAdapter() {
@Override
public void mouseClicked(MouseEvent e) {
try {
Desktop.getDesktop().browse(new URI(link));
} catch (Exception ex) {
ex.printStackTrace();
}
}
});
return label;
}

public static void main(String[] args) {
new AboutUsPage();
}
}
```

**InstructionsPage.java**

```
import javax.swing.*;
import java.awt.*;

public class InstructionsPage extends JFrame {

public InstructionsPage() {
setTitle("Instructions");
setSize(400, 400); // Adjusted size of the Instructions window
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLocationRelativeTo(null); // Center the window on the screen

JPanel panel = new JPanel();
panel.setBackground(Color.WHITE);
panel.setLayout(new BorderLayout());

// HTML content for instructions
JLabel instructionsLabel = new JLabel("<html><body style='text-align: center;'>"
+ "<h2>Flight Reservation Instructions</h2>"
+ "<ul style='text-align: left;'>"
+ "<li><b>Step 1:</b> Enter your departure and destination cities.</li>"
+ "<li><b>Step 2:</b> Select your travel dates for departure and return (if applicable).</li>"
+ "<li><b>Step 3:</b> Choose the number of passengers and cabin class (Economy, Business, or First
Class).</li>"
+ "<li><b>Step 4:</b> Review available flight options and select the one that best suits your
needs.</li>"
+ "<li><b>Step 5:</b> Enter passenger details, including full name and contact information.</li>"
+ "<li><b>Step 6:</b> Review your booking and proceed to the payment section to complete the
reservation.</li>"
+ "<li><b>Step 7:</b> After payment, you will receive a confirmation email with your booking details
and e-ticket.</li>"
+ "</ul>"
+ "<p>Note: Make sure all information is accurate to avoid issues during check-in and boarding.</p>"
+ "</body></html>");
instructionsLabel.setFont(new Font("Arial", Font.PLAIN, 14));
instructionsLabel.setHorizontalAlignment(SwingConstants.CENTER);
instructionsLabel.setVerticalAlignment(SwingConstants.CENTER);

panel.add(instructionsLabel, BorderLayout.CENTER);

// Back button
JButton backButton = new JButton("Back");
backButton.addActionListener(e -> {
new MainMenu();
dispose();
});
```

```
JPanel buttonPanel = new JPanel(new FlowLayout());
buttonPanel.add(backButton);
panel.add(buttonPanel, BorderLayout.SOUTH);

add(panel);
setVisible(true);
}

public static void main(String[] args) {
new InstructionsPage();
}
}
```

**FlightSelectionPage.java**

```
import BusinessFlightDetailsPage.BusinessFlightDetailsPage;
import javax.swing.*;
import java.awt.*;
import java.net.URL;

public class FlightSelectionPage extends JFrame {

public FlightSelectionPage() {
setTitle("AeroPass");
setSize(500, 300);
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLocationRelativeTo(null); // Center the window on the screen

// Set background image
try {
URL url = new URL("https://wallpaperaccess.com/full/254367.png"); // Replace with the URL of your
image
JLabel background = new JLabel(new ImageIcon(url));
setContentPane(new JPanel(new BorderLayout()) {
@Override
public void paintComponent(Graphics g) {
super.paintComponent(g);
g.drawImage(new ImageIcon(url).getImage(), 0, 0, getWidth(), getHeight(), this);
}
});
} catch (Exception e) {
e.printStackTrace();
}

// Add title label
JLabel title = new JLabel("The AeroPass offers two flights:", JLabel.CENTER);
title.setFont(new Font("Arial", Font.BOLD, 24));
```

```
title.setForeground(Color.WHITE); // Use dark gray text color for contrast
add(title, BorderLayout.NORTH);

// Create panel for the buttons with a transparent background
JPanel optionsPanel = new JPanel();
optionsPanel.setOpaque(false); // Transparent panel to show background image
optionsPanel.setLayout(new BoxLayout(optionsPanel, BoxLayout.Y_AXIS));

// Create buttons with light red background color and smaller size
Dimension buttonSize = new Dimension(150, 30);

JButton domesticButton = new JButton("Domestic Flights");
domesticButton.setFont(new Font("Arial", Font.PLAIN, 14));
domesticButton.setBackground(new Color(255, 182, 193)); // Light red color
domesticButton.setPreferredSize(buttonSize);
domesticButton.setMaximumSize(buttonSize);

JButton businessButton = new JButton("Business Flights");
businessButton.setFont(new Font("Arial", Font.PLAIN, 14));
businessButton.setBackground(new Color(255, 182, 193)); // Light red color
businessButton.setPreferredSize(buttonSize);
businessButton.setMaximumSize(buttonSize);

JButton backButton = new JButton("Back");
backButton.setFont(new Font("Arial", Font.PLAIN, 14));
backButton.setBackground(new Color(255, 182, 193)); // Light red color
backButton.setPreferredSize(buttonSize);
backButton.setMaximumSize(buttonSize);

// Add button actions
domesticButton.addActionListener(e -> {
new DomesticFlightDetailsPage();
dispose();
});
businessButton.addActionListener(e -> {
new BusinessFlightDetailsPage();
dispose();
});
backButton.addActionListener(e -> {
new MainMenu();
dispose();
});

// Add buttons to panel with spacing
domesticButton.setAlignmentX(Component.LEFT_ALIGNMENT);
businessButton.setAlignmentX(Component.LEFT_ALIGNMENT);
backButton.setAlignmentX(Component.LEFT_ALIGNMENT);
```

```
optionsPanel.add(Box.createVerticalStrut(10));
optionsPanel.add(domesticButton);
optionsPanel.add(Box.createVerticalStrut(10));
optionsPanel.add(businessButton);
optionsPanel.add(Box.createVerticalStrut(10));
optionsPanel.add(backButton);

// Adjust the panel to move buttons 8% to the right
JPanel buttonPanel = new JPanel(new BorderLayout());
buttonPanel.setOpaque(false);
JPanel offsetPanel = new JPanel(new FlowLayout(FlowLayout.LEFT, 30, 10)); // Adjust the insets for
8% right
offsetPanel.setOpaque(false);
offsetPanel.add(optionsPanel);
buttonPanel.add(offsetPanel, BorderLayout.WEST);

// Add the panel to the center of the layout
add(buttonPanel, BorderLayout.CENTER);

setVisible(true);
}

public static void main(String[] args) {
new FlightSelectionPage();
}
}
```

DomesticFlightDetailsPage.java

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.KeyAdapter;
import java.awt.event.KeyEvent;
import java.awt.event.MouseAdapter;
import java.awt.event.MouseEvent;

public class DomesticFlightDetailsPage extends JFrame {
private JTextField nameField, emailField, phoneNumberField, ticketsField, passportNumberField;
private JLabel priceLabel;
private boolean destinationSelected = false;
private String destinationFrom = "";
private String destinationTo = "";

public DomesticFlightDetailsPage() {
setTitle("Domestic Flight Details");
setSize(600, 400);
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLayout(new GridBagLayout());
```

```
getContentPane().setBackground(new Color(173, 216, 230)); // Light blue color

Font labelFont = new Font("Arial", Font.PLAIN, 14);
GridBagConstraints gbc = new GridBagConstraints();
gbc.insets = new Insets(10, 10, 10, 10);
gbc.anchor = GridBagConstraints.WEST;

// Name
gbc.gridx = 0;
gbc.gridy = 0;
add(createLabel("Name:", labelFont), gbc);
gbc.gridx = 1;
nameField = new JTextField(20);
add(nameField, gbc);
nameField.addKeyListener(new KeyAdapter() {
public void keyTyped(KeyEvent e) {
char c = e.getKeyChar();
if (!Character.isLetter(c) && !Character.isWhitespace(c)) {
e.consume();
}
}
});

// Email
gbc.gridx = 0;
gbc.gridy = 1;
add(createLabel("Email:", labelFont), gbc);
gbc.gridx = 1;
emailField = new JTextField(20);
add(emailField, gbc);

// Phone Number
gbc.gridx = 0;
gbc.gridy = 2;
add(createLabel("Phone Number:", labelFont), gbc);
gbc.gridx = 1;
phoneNumberField = new JTextField(20);
add(phoneNumberField, gbc);
phoneNumberField.addKeyListener(new KeyAdapter() {
public void keyTyped(KeyEvent e) {
char c = e.getKeyChar();
if (!Character.isDigit(c) || phoneNumberField.getText().length() >= 10) {
e.consume();
}
}
});
```

```
"Error", JOptionPane.ERROR_MESSAGE);
}
});
buttonPanel.add(proceedButton);

JButton backButton = new JButton("Back");
backButton.addActionListener(e -> {
new FlightSelectionPage();
dispose();
});
buttonPanel.add(backButton);
add(buttonPanel, gbc);

setVisible(true);
}

private JLabel createLabel(String text, Font font) {
JLabel label = new JLabel(text);
label.setFont(font);
return label;
}

private boolean validateFields() {
if (nameField.getText().trim().isEmpty() || emailField.getText().trim().isEmpty() ||
phoneNumberField.getText().trim().isEmpty() || ticketsField.getText().trim().isEmpty() ||
passportNumberField.getText().trim().isEmpty()) {
JOptionPane.showMessageDialog(this, "All fields must be filled", "Error",
JOptionPane.ERROR_MESSAGE);
return false;
}

if (!emailField.getText().matches("^[\\w-\\.]+@([\\w-]+\\.)+[\\w-]{2,4}$")) {
JOptionPane.showMessageDialog(this, "Invalid email format", "Error",
JOptionPane.ERROR_MESSAGE);
return false;
}

if (phoneNumberField.getText().length() != 10) {
JOptionPane.showMessageDialog(this, "Phone number must be 10 digits", "Error",
JOptionPane.ERROR_MESSAGE);
return false;
}

return true;
}

public void setDestination(String from, String to) {
this.destinationFrom = from;
```

```
this.destinationTo = to;
this.destinationSelected = true;
JOptionPane.showMessageDialog(this, "Destination selected: " + from + " to " + to, "Success",
JOptionPane.INFORMATION_MESSAGE);
}

public static void main(String[] args) {
new DomesticFlightDetailsPage();
}
}
```

**DestinationSelectionPage.java**

```
import javax.swing.*;
import java.awt.*;
import java.net.URL;

public class DestinationSelectionPage extends JFrame {
private JComboBox<String> fromComboBox, toComboBox;
private DomesticFlightDetailsPage parent;

public DestinationSelectionPage(DomesticFlightDetailsPage parent) {
this.parent = parent;

setTitle("Select Destination");
setSize(600, 400);
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLayout(new BorderLayout());

// Hide parent page while this page is open
parent.setVisible(false);

// Load background image
JLabel backgroundLabel = new JLabel();
try {
URL imageURL = new URL("https://pixnio.com/free-images/2017/05/14/2017-05-14-15-18-07.jpg");
// Replace with your image URL
ImageIcon backgroundImageIcon = new ImageIcon(imageURL);
Image backgroundImage = backgroundImageIcon.getImage().getScaledInstance(600, 400,
Image.SCALE_SMOOTH);
backgroundLabel = new JLabel(new ImageIcon(backgroundImage));
backgroundLabel.setLayout(new BorderLayout());
} catch (Exception e) {
e.printStackTrace();
}
setContentPane(backgroundLabel);

// Title label
```

```
JLabel titleLabel = createStyledLabel("Select Your Journey:");
titleLabel.setFont(new Font("Arial", Font.BOLD, 24));
titleLabel.setHorizontalAlignment(SwingConstants.CENTER);
add(titleLabel, BorderLayout.NORTH);

JPanel inputPanel = new JPanel(new GridBagLayout());
inputPanel.setOpaque(false);

GridBagConstraints gbc = new GridBagConstraints();
gbc.gridx = 0;
gbc.gridy = 0;
gbc.anchor = GridBagConstraints.WEST;

String[] places = {"Goa", "Hyderabad", "Mumbai", "Karnataka", "Kerala", "Dubai", "Gokarna",
"Maharashtra", "Chennai", "Ladakh", "Munnar", "Andhra Pradesh", "Vizag", "Arunachal Pradesh",
"America", "Australia"};
fromComboBox = new JComboBox<>(places);
toComboBox = new JComboBox<>(places);
fromComboBox.setPreferredSize(new Dimension(200, 30));
toComboBox.setPreferredSize(new Dimension(200, 30));

JPanel comboBoxPanel = new JPanel(new FlowLayout());
comboBoxPanel.setOpaque(false);
comboBoxPanel.add(fromComboBox);
comboBoxPanel.add(createStyledLabel(" to "));
comboBoxPanel.add(toComboBox);

inputPanel.add(comboBoxPanel, gbc);
add(inputPanel, BorderLayout.CENTER);

JPanel buttonPanel = new JPanel(new FlowLayout());
buttonPanel.setOpaque(false);

JButton selectDestinationButton = new JButton("Select Destination");
selectDestinationButton.setPreferredSize(new Dimension(200, 40));
selectDestinationButton.addActionListener(e -> {
String from = (String) fromComboBox.getSelectedItem();
String to = (String) toComboBox.getSelectedItem();
parent.setDestination(from, to); // Set the selected destination in parent page
parent.setVisible(true); // Show the parent page again
dispose(); // Close this page
});
buttonPanel.add(selectDestinationButton);

add(buttonPanel, BorderLayout.SOUTH);

setVisible(true);
}
```

```
private JLabel createStyledLabel(String text) {
JLabel label = new JLabel(text);
label.setFont(new Font("Arial", Font.PLAIN, 16));
label.setForeground(Color.WHITE);
return label;
}
}
```

**ParkingSlotPage.java**

```
import javax.swing.*;
import java.awt.*;

public class ParkingSlotPage extends JFrame {

private String from;
private String to;

public ParkingSlotPage(String from, String to) {
this.from = from;
this.to = to;

setTitle("Parking Slot Booking");
setSize(600, 400);
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLayout(new BorderLayout());

// Set light blue background color
Color backgroundColor = new Color(173, 216, 230);
getContentPane().setBackground(backgroundColor);

// Create the main panel
JPanel mainPanel = new JPanel(new GridLayout(3, 2, 10, 10));
mainPanel.setBackground(backgroundColor);

// Radio buttons for vehicle selection
JRadioButton twoWheelerButton = new JRadioButton("Two Wheeler");
JRadioButton fourWheelerButton = new JRadioButton("Four Wheeler");

// Set background color for radio buttons
twoWheelerButton.setBackground(backgroundColor);
fourWheelerButton.setBackground(backgroundColor);

// Add vehicle buttons to a group
ButtonGroup vehicleGroup = new ButtonGroup();
vehicleGroup.add(twoWheelerButton);
vehicleGroup.add(fourWheelerButton);
```

```
mainPanel.add(twoWheelerButton);
mainPanel.add(fourWheelerButton);

// Create the next button
JButton nextButton = new JButton("Next");
nextButton.setPreferredSize(new Dimension(100, 30));
nextButton.addActionListener(e -> {
if (twoWheelerButton.isSelected()) {
// Navigate to TwoWheelerDetailsPage (assuming this exists)
new TwoWheelerDetailsPage(from, to);
} else if (fourWheelerButton.isSelected()) {
// Navigate to FourWheelerDetailsPage (assuming this exists)
new FourWheelerDetailsPage(from, to);
}
dispose();
});

// Create the back button
JButton backButton = new JButton("Back");
backButton.setPreferredSize(new Dimension(100, 30));
backButton.addActionListener(e -> {
new DomesticFlightDetailsPage(); // Assuming this exists
dispose();
});

// Create a panel for the buttons
JPanel buttonPanel = new JPanel(new FlowLayout());
buttonPanel.setBackground(backgroundColor);
buttonPanel.add(backButton);
buttonPanel.add(nextButton);

// Add panels to the frame
add(mainPanel, BorderLayout.CENTER);
add(buttonPanel, BorderLayout.SOUTH);

setVisible(true);
}
}
```

**PriceDetailsPage.java**

```
import javax.swing.*;
import java.awt.*;

public class PriceDetailsPage extends JFrame {

public PriceDetailsPage(String from, String to, int numberOfTickets, int parkingPrice) {
setTitle("Price Details");
setSize(600, 400);
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLayout(new BorderLayout());

// Calculate the flight price based on the ticket quantity
int flightPrice = calculateFlightPrice(from, to, numberOfTickets);
int totalPrice = parkingPrice + flightPrice;

// Create and display price details
String priceDetailsText = "<html>Parking Amount: " + parkingPrice + "<br>" +
"Flight Amount: " + flightPrice + "<br>" +
"Total Amount: " + totalPrice + "</html>";
JLabel priceDetailsLabel = new JLabel(priceDetailsText, JLabel.CENTER);
priceDetailsLabel.setFont(new Font("Arial", Font.PLAIN, 16));
add(priceDetailsLabel, BorderLayout.CENTER);

// Proceed to payment button
JButton proceedButton = new JButton("Proceed to Payment");
proceedButton.setPreferredSize(new Dimension(150, 40));
proceedButton.addActionListener(e -> {
new PaymentGatewayPage(totalPrice); // Pass the total price to PaymentGatewayPage
dispose(); // Close the PriceDetailsPage
});
add(proceedButton, BorderLayout.SOUTH);

// Set the frame visible
setVisible(true);
}

PriceDetailsPage(String from, String to, int parkingPrice) {
throw new UnsupportedOperationException("Not supported yet."); // Generated from
nbfs://nbhost/SystemFileSystem/Templates/Classes/Code/GeneratedMethodBody
}

// This method calculates the flight price based on the number of tickets
private int calculateFlightPrice(String from, String to, int numberOfTickets) {
// Example price per ticket
int pricePerTicket = 1000;
```

```
return pricePerTicket * numberOfTickets;
}

public static void main(String[] args) {
// Example to open the price details page with sample data
new PriceDetailsPage("New York", "Los Angeles", 1, 200); // 1 ticket, 200 parking fee
}
}
```

**TwoWheelerDetailsPage.java**

```
import javax.swing.*;
import java.awt.*;

public class TwoWheelerDetailsPage extends JFrame {

private JTextField vehicleNumberField, modelField, hoursField;
private JLabel priceLabel;
private String from, to;

public TwoWheelerDetailsPage(String from, String to) {
this.from = from;
this.to = to;

setTitle("Two Wheeler Details");
setSize(600, 400);
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLayout(new GridBagLayout());

getContentPane().setBackground(new Color(173, 216, 230)); // Light blue color

GridBagConstraints gbc = new GridBagConstraints();
gbc.insets = new Insets(10, 10, 10, 10); // Padding around components
gbc.anchor = GridBagConstraints.WEST;

// Add labels and text fields
gbc.gridx = 0;
gbc.gridy = 0;
add(createLabel("Vehicle Number:"), gbc);
gbc.gridx = 1;
vehicleNumberField = new JTextField(15);
add(vehicleNumberField, gbc);

gbc.gridx = 0;
gbc.gridy = 1;
add(createLabel("Model:"), gbc);
gbc.gridx = 1;
modelField = new JTextField(15);
```

```
add(modelField, gbc);

gbc.gridx = 0;
gbc.gridy = 2;
add(createLabel("How many hours:"), gbc);
gbc.gridx = 1;
hoursField = new JTextField(15);
add(hoursField, gbc);

gbc.gridx = 0;
gbc.gridy = 3;
add(createLabel("Price:"), gbc);
gbc.gridx = 1;
priceLabel = new JLabel();
add(priceLabel, gbc);

JPanel buttonPanel = new JPanel(new FlowLayout());
buttonPanel.setOpaque(false);

JButton dontNeedButton = new JButton("Don't Need");
dontNeedButton.setPreferredSize(new Dimension(100, 30));
dontNeedButton.addActionListener(e -> {
new ParkingSlotPage(from, to);
dispose();
});
buttonPanel.add(dontNeedButton);

JButton okButton = new JButton("OK");
okButton.setPreferredSize(new Dimension(100, 30));
okButton.addActionListener(e -> {
int hours;
try {
hours = Integer.parseInt(hoursField.getText());
if (hours > 5) {
JOptionPane.showMessageDialog(this, "No, you are not allowed to park for more than 5 hours",
"Error", JOptionPane.ERROR_MESSAGE);
} else {
int price = hours * 100; // Calculate the price based on hours
priceLabel.setText(String.valueOf(price));
JOptionPane.showMessageDialog(this, "Ok, you can proceed. Price: " + price, "Information",
JOptionPane.INFORMATION_MESSAGE);
new LuggageTransportPage(from, to, price).setVisible(true);
dispose();
}
} catch (NumberFormatException ex) {
JOptionPane.showMessageDialog(this, "Please enter a valid number of hours", "Error",
JOptionPane.ERROR_MESSAGE);
}
```

```
});
buttonPanel.add(okButton);

gbc.gridx = 0;
gbc.gridy = 4;
gbc.gridwidth = 2;
gbc.anchor = GridBagConstraints.CENTER;
add(buttonPanel, gbc);

setVisible(true);
}

private JLabel createLabel(String text) {
JLabel label = new JLabel(text);
label.setFont(new Font("Arial", Font.PLAIN, 14));
return label;
}

public static void main(String[] args) {
new TwoWheelerDetailsPage("defaultFrom", "defaultTo");
}
}
```

**FeedbackPage.java**

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.awt.event.KeyAdapter;
import java.awt.event.KeyEvent;
import java.net.URL;

class FeedbackPage extends JFrame {
public FeedbackPage() {
setTitle("Feedback");
setSize(500, 400);
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setLayout(new BorderLayout());

// Load background image
JLabel backgroundLabel = new JLabel();
try {
URL imageURL = new URL("https://media.istockphoto.com/id/1306480742/vector/user-reviews-and-feedback-concept.jpg?s=612x612&w=0&k=20&c=7Dcc3LW5sAhXUKKUJWB2JMgSuFWI4ebV-Ac8lc4Wjno="); // Replace with your image URL
ImageIcon backgroundImageIcon = new ImageIcon(imageURL);
Image backgroundImage = backgroundImageIcon.getImage().getScaledInstance(500, 400,
```

```
Image.SCALE_SMOOTH);
backgroundLabel = new JLabel(new ImageIcon(backgroundImage));
backgroundLabel.setLayout(new BorderLayout());
} catch (Exception e) {
e.printStackTrace();
}
setContentPane(backgroundLabel);

JLabel titleLabel = new JLabel("We value your feedback!", JLabel.CENTER);
titleLabel.setFont(new Font("Arial", Font.BOLD, 24));
titleLabel.setForeground(Color.BLACK); // Set the title text color to black
add(titleLabel, BorderLayout.NORTH);

JPanel feedbackPanel = new JPanel();
feedbackPanel.setLayout(new GridLayout(5, 2, 10, 10));
feedbackPanel.setOpaque(false); // Make the panel background transparent

// Name field - only allow letters
feedbackPanel.add(new JLabel("Name:"));
JTextField nameField = new JTextField(20);
nameField.addKeyListener(new KeyAdapter() {
@Override
public void keyTyped(KeyEvent e) {
char c = e.getKeyChar();
if (!Character.isLetter(c) && !Character.isWhitespace(c) && c != KeyEvent.VK_BACK_SPACE) {
e.consume(); // Ignore non-alphabetic input
}
}
});
feedbackPanel.add(nameField);

// Email field - simple validation
feedbackPanel.add(new JLabel("Email:"));
JTextField emailField = new JTextField(20);
emailField.addKeyListener(new KeyAdapter() {
@Override
public void keyReleased(KeyEvent e) {
String email = emailField.getText();
if (!email.matches("^[\\w-\\.]+@([\\w-]+\\.)+[\\w-]{2,4}$")) {
emailField.setForeground(Color.RED); // Invalid email
} else {
emailField.setForeground(Color.BLACK); // Valid email
}
}
});
feedbackPanel.add(emailField);

// Rating field
```

```
feedbackPanel.add(new JLabel("Rating (1-5):"));
JComboBox<String> ratingComboBox = new JComboBox<>(new String[]{"1", "2", "3", "4", "5"});
feedbackPanel.add(ratingComboBox);

// Comments field - only allow alphabetic input
feedbackPanel.add(new JLabel("Comments:"));
JTextArea commentsArea = new JTextArea(3, 20);
commentsArea.setLineWrap(true);
commentsArea.setWrapStyleWord(true);
commentsArea.addKeyListener(new KeyAdapter() {
@Override
public void keyTyped(KeyEvent e) {
char c = e.getKeyChar();
if (!Character.isLetter(c) && c != ' ' && c != KeyEvent.VK_BACK_SPACE) {
e.consume(); // Ignore non-alphabetic characters
}
}
});
feedbackPanel.add(new JScrollPane(commentsArea));

add(feedbackPanel, BorderLayout.CENTER);

// Submit button
JButton submitButton = new JButton("Submit Feedback");
submitButton.addActionListener(new ActionListener() {
@Override
public void actionPerformed(ActionEvent e) {
String name = nameField.getText();
String email = emailField.getText();
String rating = (String) ratingComboBox.getSelectedItem();
String comments = commentsArea.getText();

// Validate inputs
if (name.isEmpty() || email.isEmpty() || rating == null || comments.isEmpty()) {
JOptionPane.showMessageDialog(FeedbackPage.this,
"All fields are required!",
"Validation Error",
JOptionPane.ERROR_MESSAGE);
return;
}

// Simple email validation
if (!email.matches("^[\\w-\\.]+@([\\w-]+\\.)+[\\w-]{2,4}$")) {
JOptionPane.showMessageDialog(FeedbackPage.this,
"Please enter a valid email address!",
"Validation Error",
JOptionPane.ERROR_MESSAGE);
return;
```

```
}

// Show a message confirming feedback submission
JOptionPane.showMessageDialog(FeedbackPage.this,
"Thank you, " + name + ", for your feedback!",
"Feedback Submitted",
JOptionPane.INFORMATION_MESSAGE);

// You can add logic here to store feedback in a file or database

// Close the feedback page after submission
dispose();
}
});

add(submitButton, BorderLayout.SOUTH);

setVisible(true);
}

public static void main(String[] args) {
// For testing, create a FeedbackPage instance
new FeedbackPage();
}
}
```

**DatabaseConnection.java**

```
package DatabaseConnection;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DatabaseConnection {

public static Connection getConnection() {
String url = "jdbc:mysql://localhost:3306/airline";  // Ensure your database name is correct
String user = "root";  // Your MySQL username
String password = "2005";  // Your MySQL password



try {
return DriverManager.getConnection(url, user, password);
} catch (SQLException e) {
e.printStackTrace();
}
```

## 5.1 Screenshots of Application

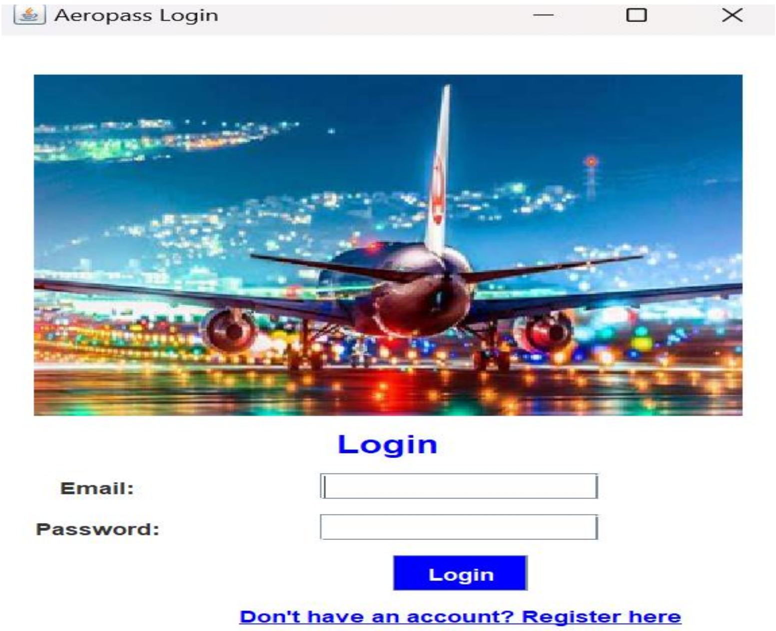



**Fig2.1 Login Page**


**Fig2.2 Regestration page**

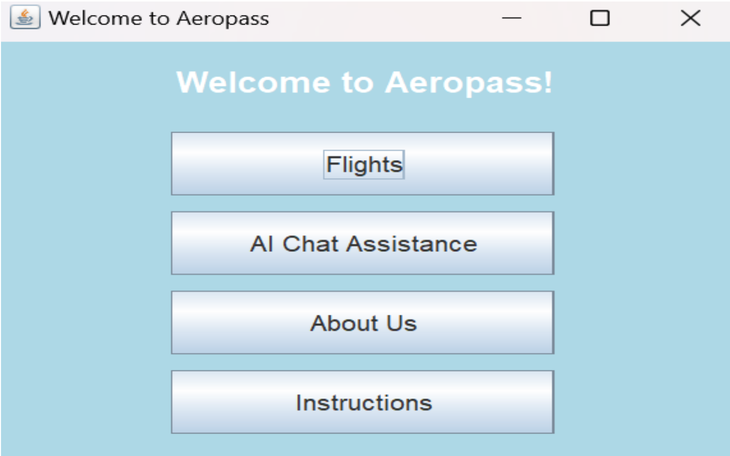



**Fig2.3 MainMenu Page**

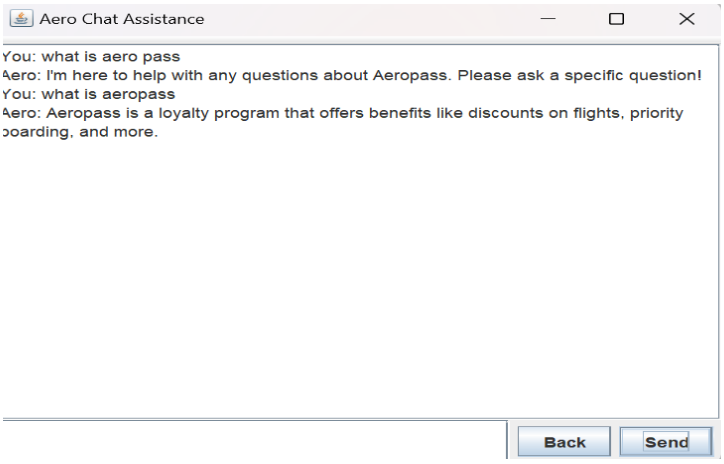



**Fig.2.4 Aero Chat Assistance**

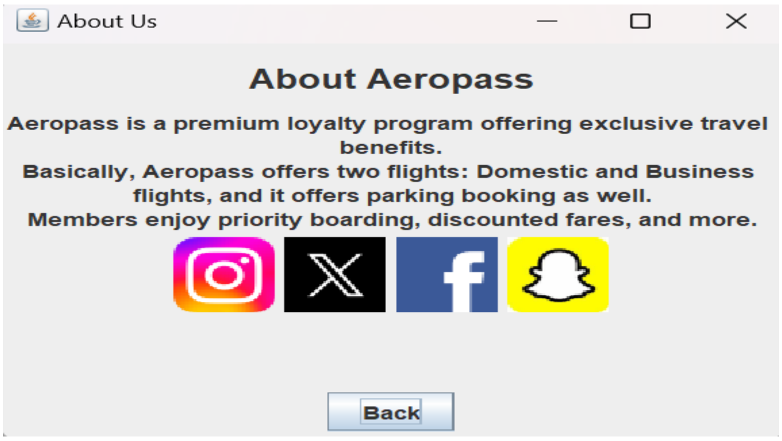



**Fig.2.5 About Us page**

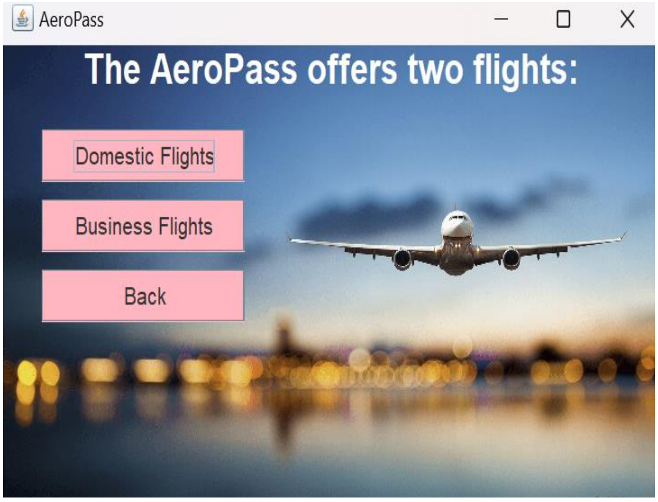



**Fig.2.6 FlightSelection Page**

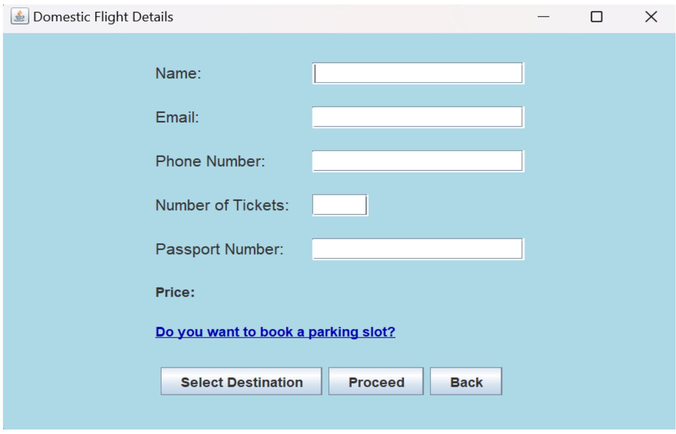



**Fig.2.7 Domestic Flight Details**

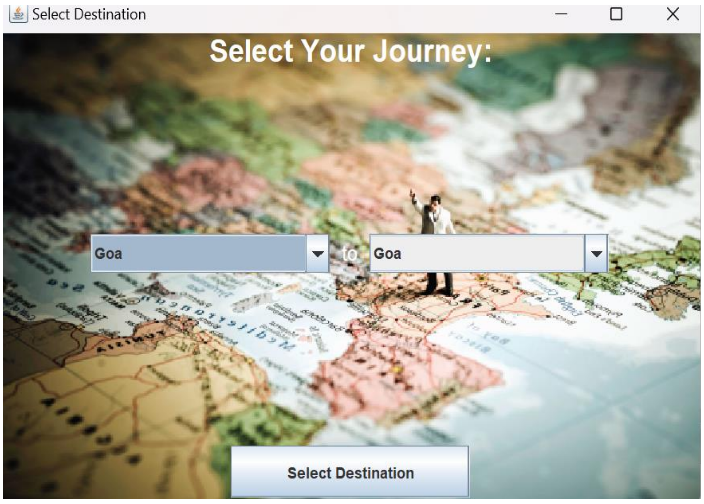



**Fig.2.8 DestinationSelection Page**

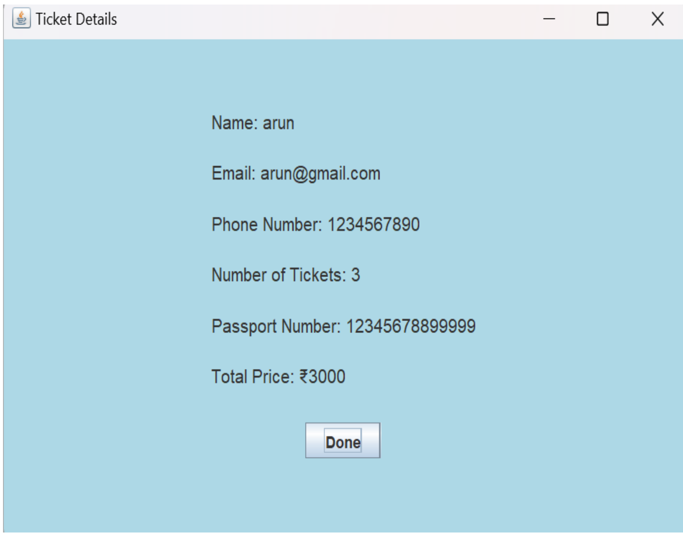



**Fig.2.9 Ticket Details Page**


**Fig.2.10 ParkingSlot Page**

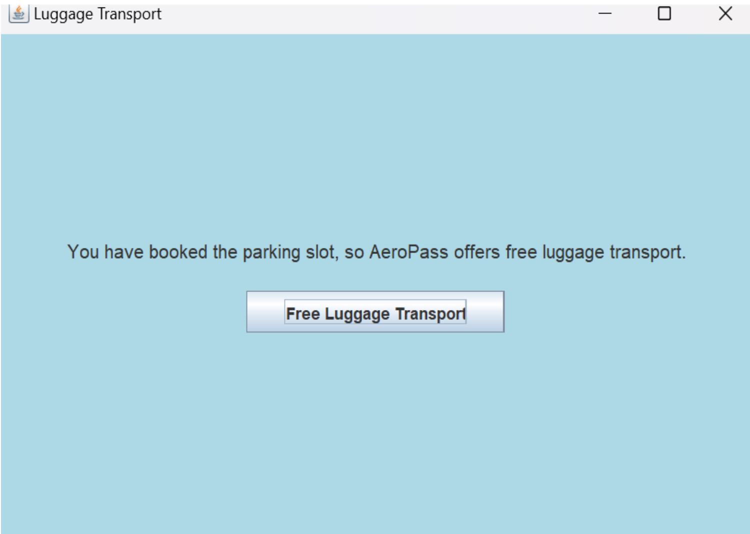



**Fig.2.11 Luggage Transport Page**

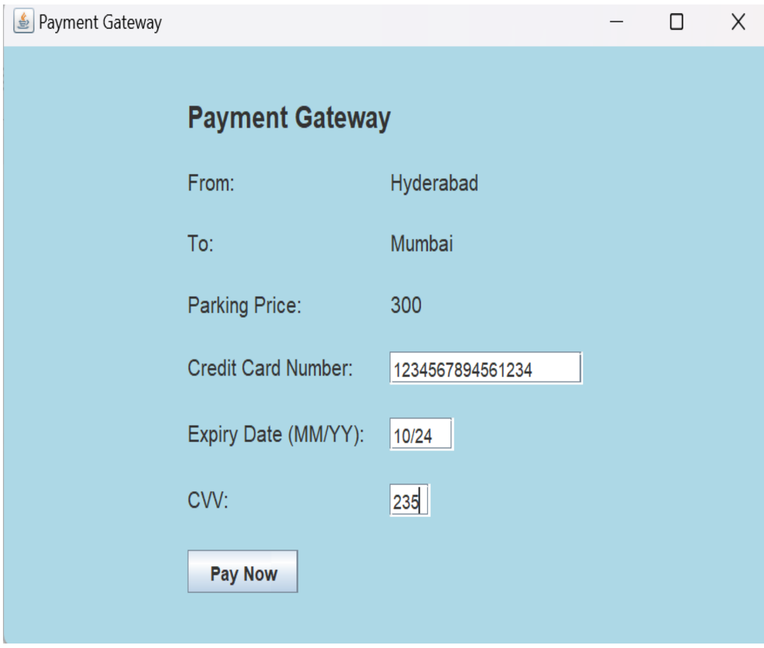



**Fig.2.12 Payment Gateway for parking**

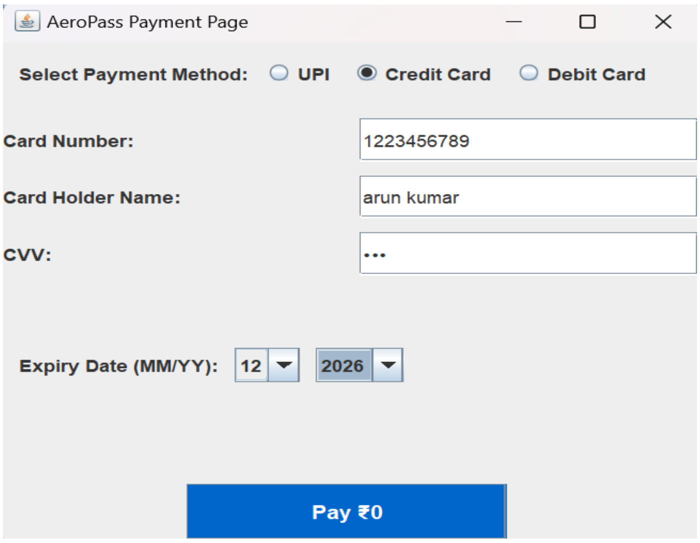



**Fig.2.13 Payment Gateway page**

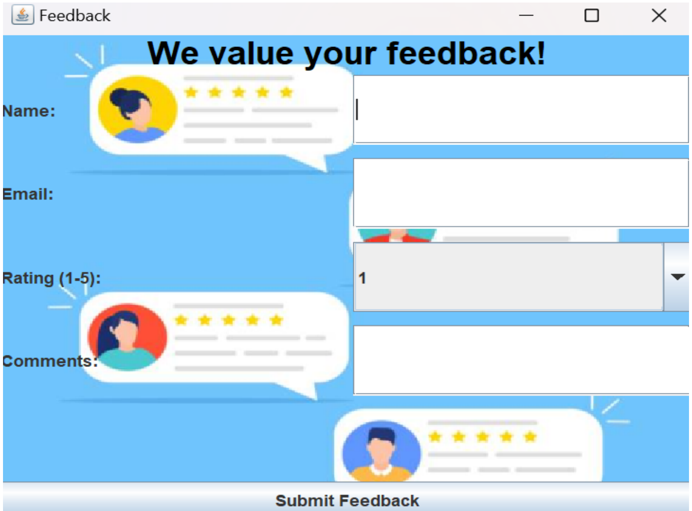



**Fig.2.14 Feedback page**

# CHAPTER-6

# Result & Conclusion

## 6.1 Result

The Aeropass airline reservation system has redefined the travel experience by combining convenience, efficiency, and innovative features. Its intuitive interface and real-time flight availability enable passengers to easily search for flights, compare prices, and complete bookings in just a few clicks, significantly reducing the time and effort required for trip planning. Additionally, features like parking assistance, which allows users to pre-book airport parking spaces, and free luggage assistance for seamless baggage handling at check-in, address common travel challenges and elevate the overall passenger experience.

For airline operators, Aeropass delivers operational efficiency by automating processes such as seat allocation, booking tracking, and real-time schedule integration. This automation minimizes errors, reduces administrative workloads, and allows airlines to quickly adapt to booking trends, cancellations, and operational changes. Furthermore, integrating auxiliary services like parking reservations and luggage tracking enhances operational workflows and provides a unified system for managing additional passenger needs. These improvements not only streamline internal processes but also create opportunities for airlines to offer value-added services and enhance customer satisfaction.

Security remains a top priority for Aeropass, ensuring the protection of sensitive user data and financial transactions through robust measures like SSL/TLS encryption, secure authentication protocols such as OAuth2, and PCI-DSS compliant payment gateways. These security protocols extend to auxiliary features, safeguarding data for services like parking and luggage assistance. By combining a seamless booking experience, operational excellence, and robust security, Aeropass has set a new benchmark in modern airline reservation systems, improving travel convenience and efficiency for passengers while addressing the operational needs of airlines.

## 6.2 Conclusion

The **Aeropass** airline reservation system successfully addresses key challenges faced by both passengers and airlines. By offering a user-friendly interface, real-time flight information, and a streamlined booking process, **Aeropass** significantly enhances the passenger experience. For airlines, the system automates critical operations, reduces manual errors, and improves efficiency, leading to smoother management of bookings and better resource allocation.

The system's scalable architecture ensures it can handle varying traffic loads, making it adaptable to both low and high-demand periods. Robust security features guarantee the protection of user data and financial transactions, fostering trust among passengers and operators alike. With its real-time data integration and insightful reporting tools, **Aeropass** also empowers airlines to make data-driven decisions, optimize pricing strategies, and improve overall business performance.

As the travel industry continues to evolve, **Aeropass** provides a forward-thinking solution that not only meets current demands but is also adaptable to future technological advancements. The system's integration with external APIs for flight data and payment processing ensures that it remains flexible and capable of incorporating new features and updates as required by both airlines and passengers.

In conclusion, **Aeropass** represents a significant step forward in airline reservation technology. It improves efficiency, enhances user satisfaction, and provides airlines with a powerful tool to manage their operations more effectively. By offering a reliable, secure, and scalable solution, **Aeropass** has the potential to redefine the way airlines manage reservations and interact with customers, positioning them for long-term success in a competitive market.

# REFERENCES

[1] I. Androutsopoulos, G.D. Ritchie, and P. Thanisch, Natural Language Interfaces to Databases – AnIntroduction, Journal of Engineering 1 Part 1 (1995), 29–81. Natural Language

[2] T. Johnson. Natural Language Computing: The Commercial Applications. Ovum Ltd., London, 1985

[3] Woods, W.A. "Progress in Natural Language Understanding: An Application to Lunar Geology", AFIPS Conf.Proc. 42, pp. 141-450, 1973

[4] D.L. Waltz. An English Language Question Answering System for a Large Relational Database. Communications of the ACM, 21(7):526–539, July 1978.

[5] G. Hendrix, E. Sacerdoti, D. Sagalowicz, and J. Slocum. Developing a Natural Language Interface to Complex Data. ACM Transactions on Database Systems, 3(2):105–147, 1978.

[6] P. Auxerre. MASQUE Modular Answering System for Queries in English - Programmer‟s Manual. Technical Report AIAI/SR/11, Artificial Intelligence Applications Institute, University of Edinburgh, March 1986.

[7] Daniel Jurafsky and James H. Martin, Speech and Language Processing: An Introduction to Natural Language Processing, Speech Recognition, and Computational Linguistics. 2nd edition. Prentice-Hall, 2009